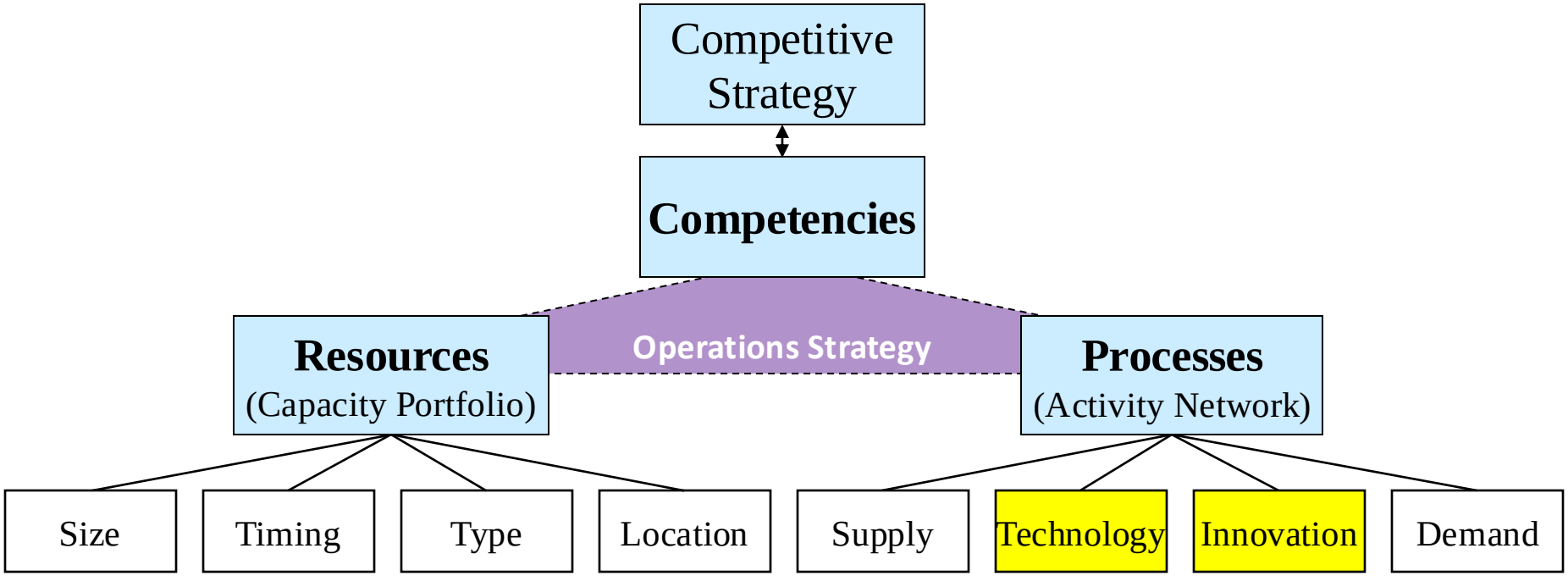
Operations Strategy Framework



Change is the only constant:

- Customers
- Competitors
- Technology

Improvement

Organization Learning

Today's Learning objectives

Be able to

- articulate product and process technology and their strategic roles in operations strategy
- explain the process of improvement and its role in operations strategy
  - Building capabilities for improvement and innovation must be an integral part of the theory and practice of operations strategy
- identify different types of learning and their impacts on improvement and innovation
  - Conceptual and operational learning
  - Learning before doing and learning by doing
- apply the learning curve as a tool to predict aggregate cost dynamics
  - The learning curve is an empirical observation of changing input requirements (labor hour or costs) as a function of experience (volume or time or other dynamics of learning)

Product Technology Examples

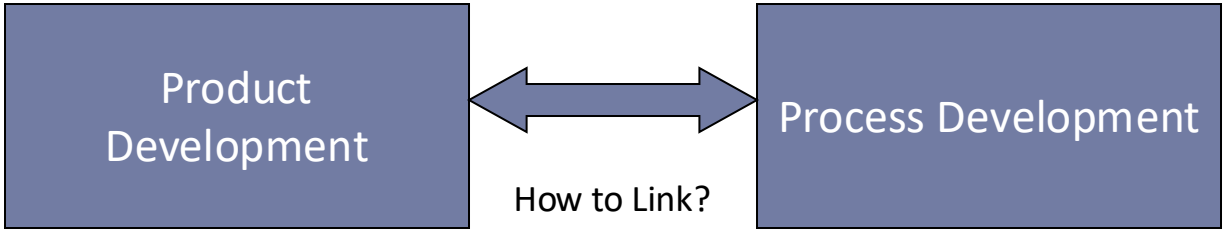
- Quantum mechanics is necessary to the understanding and designing of modern electronic devices
  - Transistor (e.g., Bulova electrochromic watch was one of the first consumer products to use a transistor)
  - Electron microscope (often used for quality control and failure analysis)
  - Flash memory chips found in USB drives (uses quantum tunnelling to erase their memory cells)
  - …
- Relate to our Cases
  - Consumer electronics
    - SWISS Watch
    - ACC/DJC connectors
  - Biotech and Pharmaceutical companies
    - Eli Lily
    - Moderna

Process Technology Examples

- Manufacturing technology
- Transportation technology
- Coordination technology
  - **GPS** exploits the principles of quantum mechanics as well as those of Special and General Relativity which is a direct application of microwave atomic clock
    - Triangulation with signals from synchronized atomic clocks embarked on a swarm of satellites circling the Earth. Precision is about 1 meter.
    - Atomic clocks combine a quartz crystal oscillator with an ensemble of atoms to achieve greater stability. NASA's Deep Space Atomic Clock will be off by less than a nanosecond after four days and less than a microsecond (one millionth of a second) after 10 years. Atomic clocks based on locking a laser on an optical transition are recent application (2017) of the frequency stability and long coherence time of Lasers. A clock at 4.1014 Hz with an uncertainty less than 1 second in age of the universe!
  - Relate to HBS cases: watch, connectors, drugs and vaccines
  - DAO2703: CAD, CAM, CAPP, NC, CNC, AGV, AS/RS , CAPP, CIM, CNC machines, IoT, FMS, Drones, robots etc…

Process technology decisions

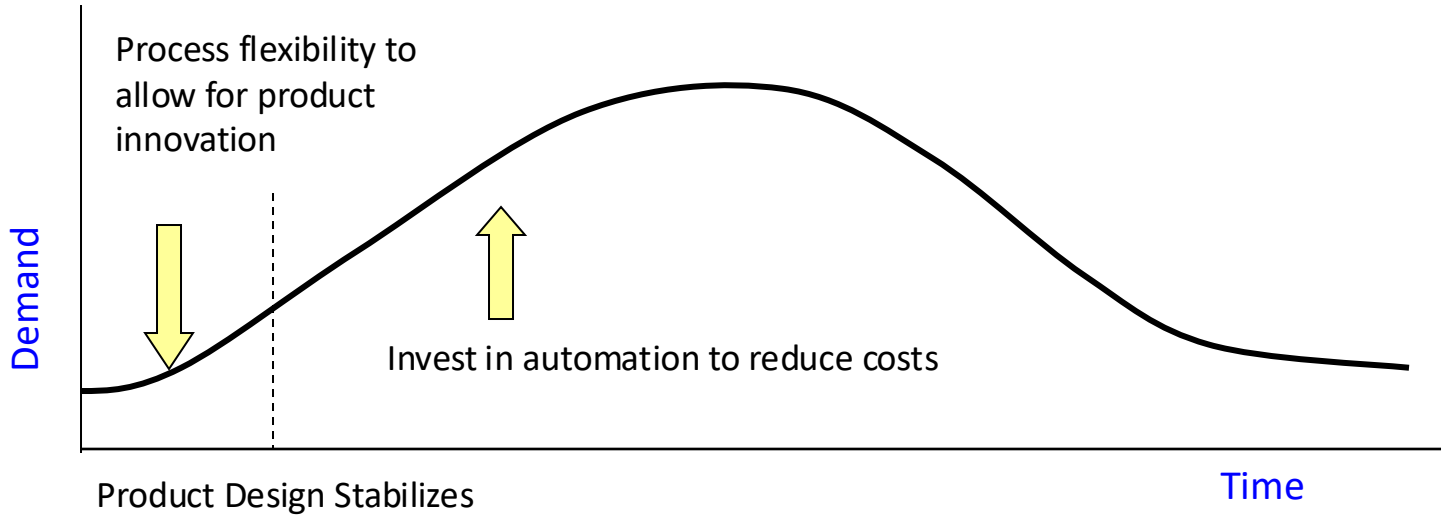
- Link with product technology and product development



- How to integrate with product development?
  - Simultaneous (or concurrent engineering)
  - Design for manufacturing (product design rules)
  - Design for Six Sigma
  - Manufacturing's involvement in prototyping
- What is at stake?
  - time
  - costs and yield
  - conformance quality
  - volume (Pandemic vs. endemic)

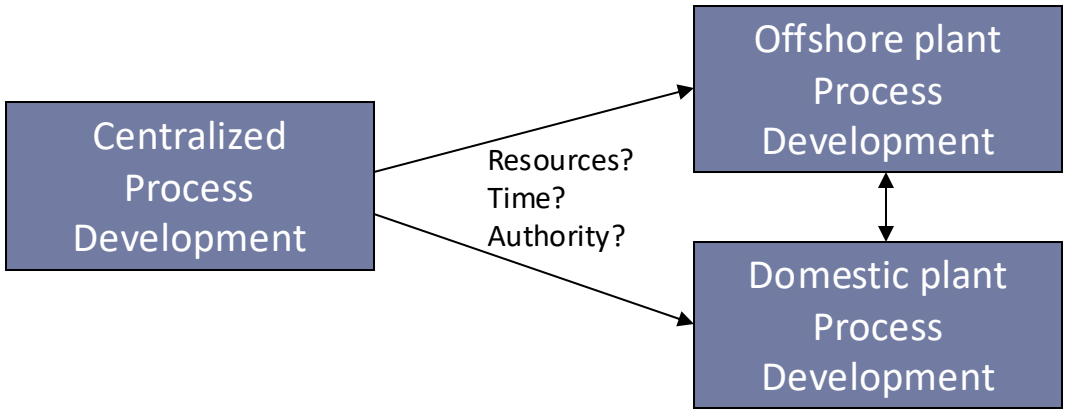
Process technology decisions

- match with the stage of the product (technology) life cycle
  - Automated, connected, and dedicated processes
    - are less flexible but this is less problematic as product designs stabilize
    - can deliver lower variable costs and thus lower average costs as production volumes increase



Process technology decisions

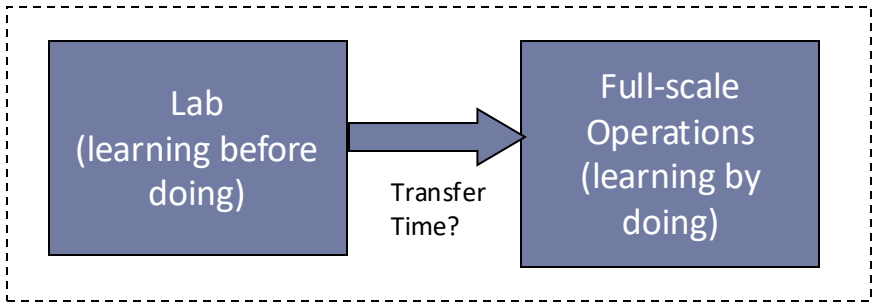
- Centralized or decentralized development?

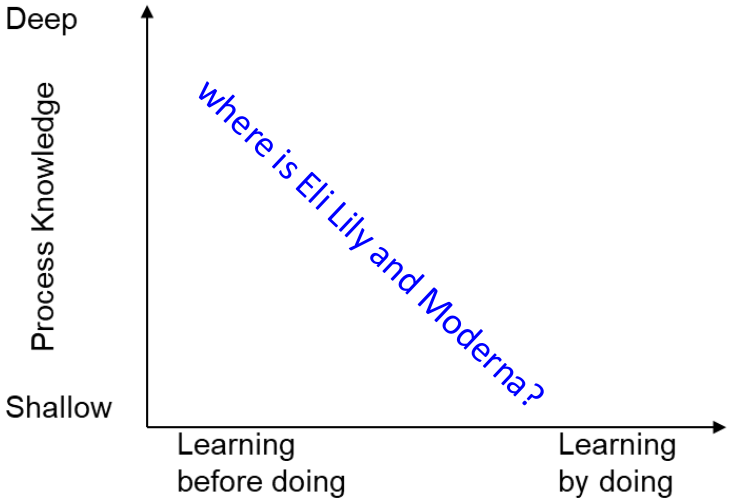


- Centralization and the enabled standardization
  - can be important for global customers
  - can lead to efficiencies (shared learning, less duplication, ability to move production load around)
  - can make it more difficult to adapt to local input factors and output requirements

Process technology decisions

- Learning before doing or learning by doing?





- Timing of transfer depends on the
  1. Depth of process knowledge
  2. Fidelity of experiments (ability of “lab” to simulate manufacturing scale behavior)
  3. Cost of experiments (lab versus full-scale)

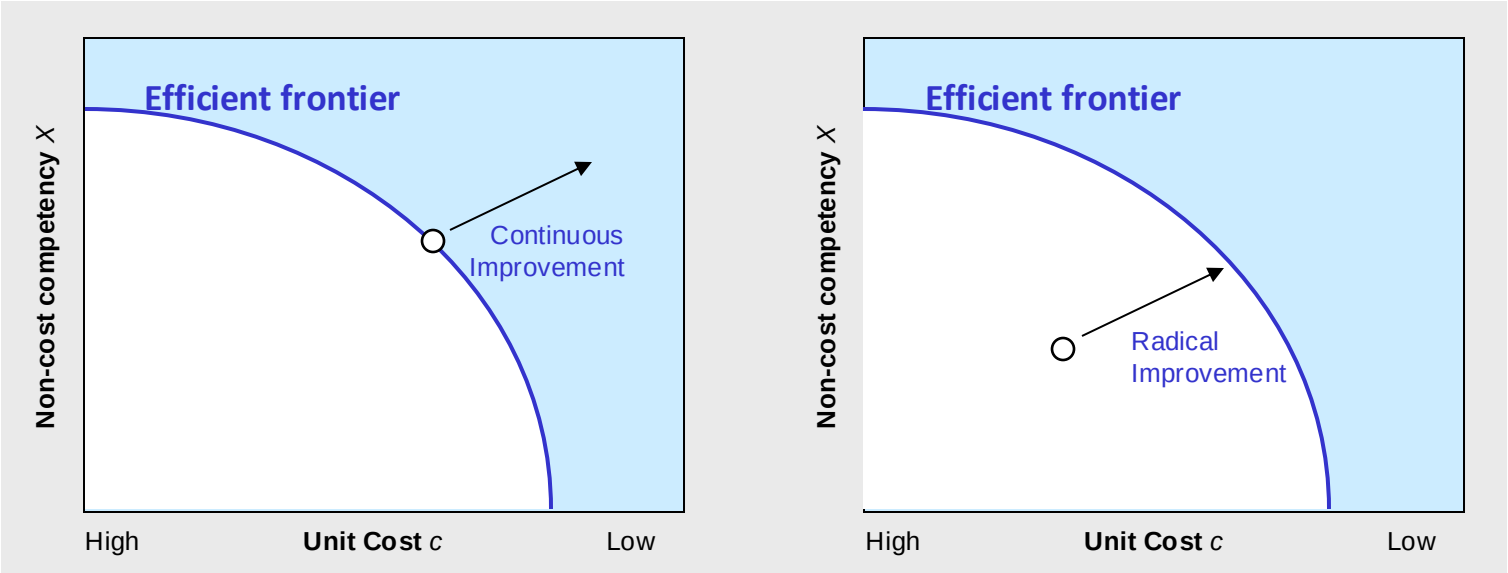
Innovation vs. Improvement

- Innovation is about creating something new
  - That can be both significant and incremental that may or may not accumulate to radical change
- Improvements means making something better
  - That can be both radical (reengineering) and continuously (Lean and Six Sigma)

💁‍♀️ Both are <u>processes</u> that result in change; difference is in the implied extend of the change and the time scale on which change occurs

Both must be an integral part of OS

- Building capabilities for improvement and innovation must be an integral part of the theory and practice of operations strategy
- Fundamentals in OS is how to stay ahead and adapt and align competences to changing customer preferences
- Two improvement strategies: Radical or Continuous

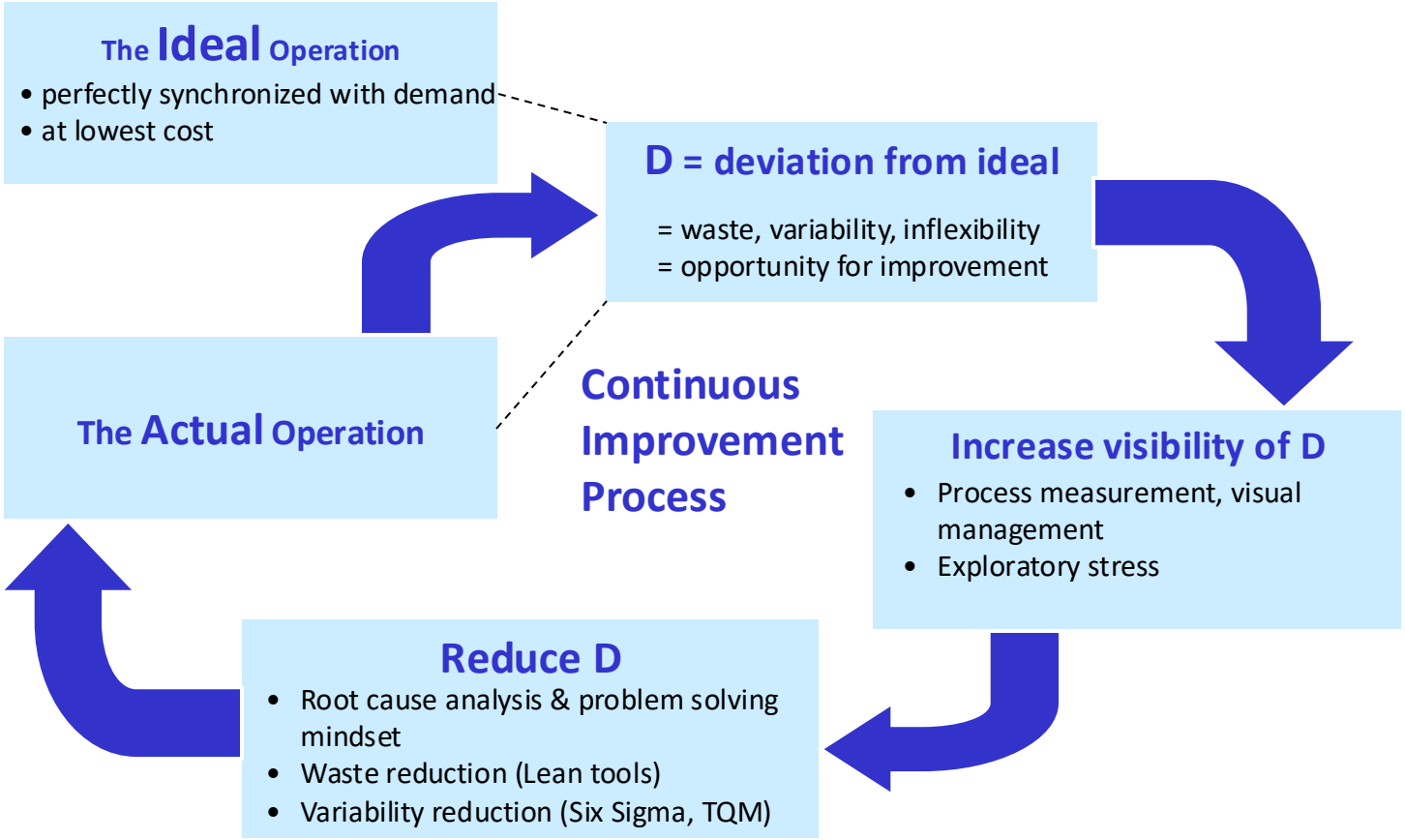


Continuous Improvement (CI)

- Continuous Improvement when seeking steady, incremental change – projects are typically <u>low risk</u>
- The following principles guide continuous improvement projects
  - Any improvement must be made in accordance with the scientific method: generate hypotheses, test, then implement
  - Process variability is reduced to eliminate problems - all work is highly specified as to content, sequence, timing and outcome
  - All employees are empowered to make improvements; managers are coaches
  - All activities, connections and flow paths have built-in tests to signal problems automatically (= control)

Adapted from: Spears, The DNA of the Toyota Production System, Harvard Business Review, 1999

Continuous Improvement as a process



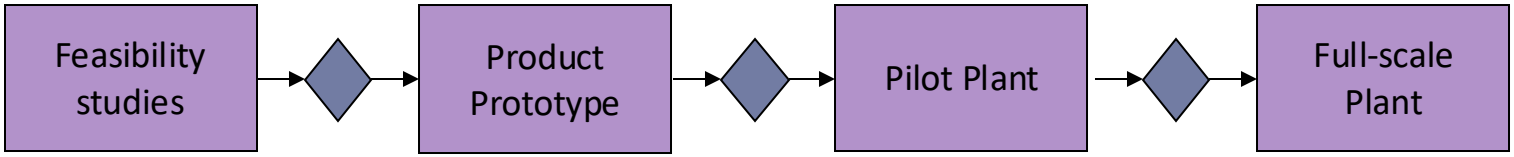
Copy Best Practice

- Trying to copy an active best practice representing a large, breakthrough improvement (e.g., delayed differentiation, internet store, etc.) – these are large projects with medium risk
- Follow these guidelines when copying best practice
  - Make sure the system can be copied and is worth copying
  - Work from a single, active template (process you want to copy)
  - Copy as closely and completely as you can (plant & equipment, skill sets and procedures)
  - Don't reformulate, adapt, or improve upon until you have reached acceptable results
  - When problems occur, refer-back to the active template
- Be aware that
  - Neither people nor documents can fully explain how a complex system works
  - We are often overconfident about our understanding of why something works

Adapted from : Szulanski & Winter, 2002. "Getting It Right the Second Time", Harvard Business Review

Reengineering

- Reengineering when trying to achieve breakthrough, superior performance with a new, unproven process – these are potentially large projects with <u>high risk</u>
- Some of the typical guidelines for those projects are to:
  - Identify the knowledge gaps and learn proactively
  - Use a stage-gate process to start small and reduce risk quickly before committing large amounts of resources

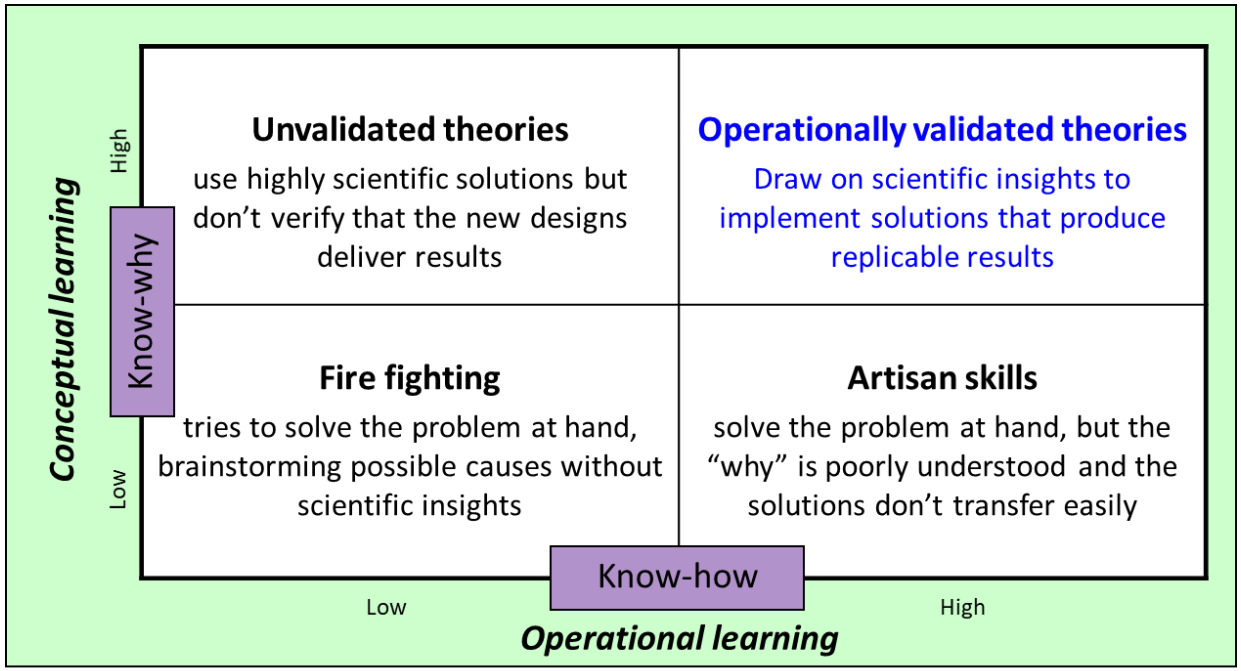


  - Use simulation to test new products/processes and avoid errors
  - Use partners/contractors to manage risk

Sources:
- Loch, C.H., De Meyer, A. and Pich, M.T. (2006) Managing the unknown: a new approach to managing high uncertainty and risk in projects. London: J. Wiley and Sons
- Hammer, M., J. Champy. 1993. Reengineering the Corporation. HarperCollins, New York.

Learning and Improvement

- Organizational learning is directly related to both improvement and innovation, the literature breaks down organization learning processes into two activities: conceptual learning and operational learning



Source: Learning across Lines: the secret to more efficient factories (HBR 2002)

Learning and Improvement

- Organizational learning is directly related to both improvement and innovation, there are two modes of learning
  - Learning before doing
    - Unleashes the power of science and technology to improve
    - Encourages a safe and efficient organization that promotes knowledge sharing and ensures that complex processes will work
    - Is compatible with step change and even drastic improvement
  - Learning by doing
    - Unleashes creativity of the whole workforce and its power to improve
    - Encourages a dynamic organization that minimizes planning time
    - Is compatible with a continuous improvement process and essential to Toyota's *Kaizen* system

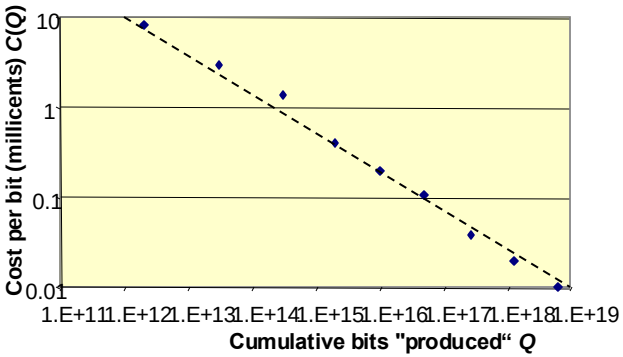
The Learning Curve

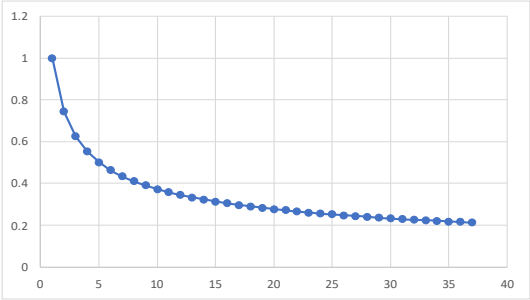
- People engaged in repetitive tasks learns by doing. If data of this phenomenon is gathered, a learning curve is found
  - The learning curve shows how learning by doing translate into improvement
- Empirically derived learning curves indicate "learning by doing is driven by cumulative volume"
  - We learn each time we process a unit, and the learning cumulates
  - In its strictest interpretation, it depicts the relationship between direct labor hours per unit and the cumulative number of units produced
  - A striking observation is that the data pretty much defines a linear relationship between direct labor hour per unit processed against cumulative production volume if using log-log scale, i.e., Log (marginal direct labor hour) vs. Log (cumulative volume) graph appears as a straight line
  - indicating that the amount of direct labor hours needed for one unit of production reduces by a similar percentage every time the cumulative volume doubles, and this percentage is often called the learning rate
  - equivalently, each doubling of cumulative volume decreases labor hour by the same learning rate

Moore's Law and the learning curve

- Moore's law (Gordon Moore 1964) is the observation that the number of transistors in a dense integrated circuit (IC) doubles about every two years. It is an observation and projection of a historical trend. Rather than a law of physics, it is an empirical relationship linked to gains from experience in production.
- Intel's Robert Noyce said in 1977 that "between 1960 and 1977 annual usage of transistors has increased by 2000 times, or has doubled 11 times, in the past 17 years. This stunning increase promotes continual cost reductions of 28% for each doubling of volume." [*Scientific American*, 1977]
- Estimate Learning: $\log C(Q) = \log C_1 - \lambda \log Q$
  - C_1 is the marginal cost of first unit
  - λ is the slope measured on a log-log scale (= 3/7 from right lower graph)
- This is Equivalent to: $C(Q) = C_1 Q^{-\lambda}$, indicating
  - $C(Q) - C(2Q) = (1- 2^{-\lambda})C(Q)$ *and*
  - <u>The Learning rate</u> is $1- 2^{-\lambda}$ =25.7% (=$1-2^{-3/7}$)

Year	Bits/die	Line width (micron)	Cost (millicents/bit)	Cumulative bits sold
1974	4.1E+03	6	8.02	2.00E+12
1977	1.6E+04	3.5	2.89	3.00E+13
1980	6.6E+04	2.3	1.37	3.00E+14
1983	2.6E+05	1.3	0.395	2.00E+15
1986	1.1E+06	1	0.196	1.00E+16
1989	4.2E+06	0.8	0.105	5.00E+16
1992	1.7E+07	0.5	0.038	2.70E+17
1995	6.7E+07	0.35	0.019	1.25E+18
1998	2.7E+08	0.25	0.01	6.25E+18





The Learning Curve (Cont.)

- Learning is, in a certain sense, “predictable”
  - means that we can extrapolate into the future with reasonable accuracy
- Learning curve is useful for
  - Capacity Planning & budgeting of resources
  - To check whether implementation of new processes “is on track”
- Other learning models (that also have empirical support) indicate learning is driven by *time, and* other “*dynamics*” of learning
  - These models stipulate annual productivity growths, annual reductions in defect levels, etc.
- Don’t be fooled though. Learning and improvement does not “just happen”

Put it together

Main topic = prepare for the future = build competencies/capabilities for innovation and improvement

↓

Learning

By doing
= process improvement

↓

Learning curve
kaizen & standardize work

↓

e.g., Cost ↓

Before doing
= product or process innovation

↓

Design strategies
Plan, Iterate, or Select strategies

↓

e.g., Functionality ↑

Interplay between these two learning modes *predicts*
Improvement and innovation dynamics